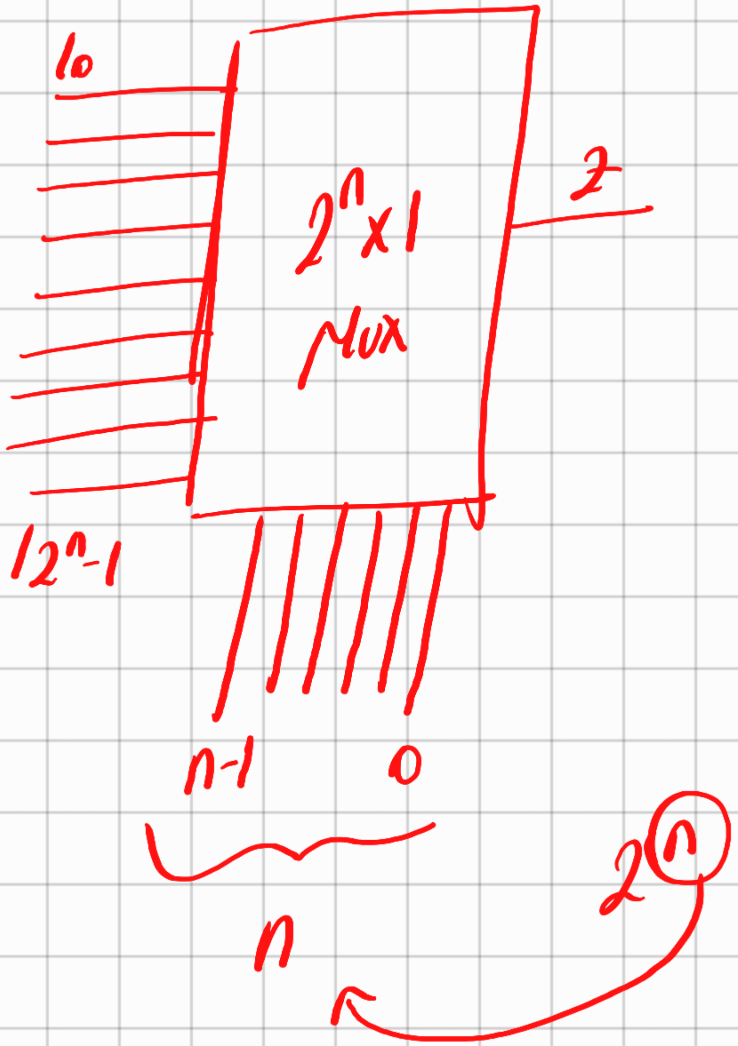
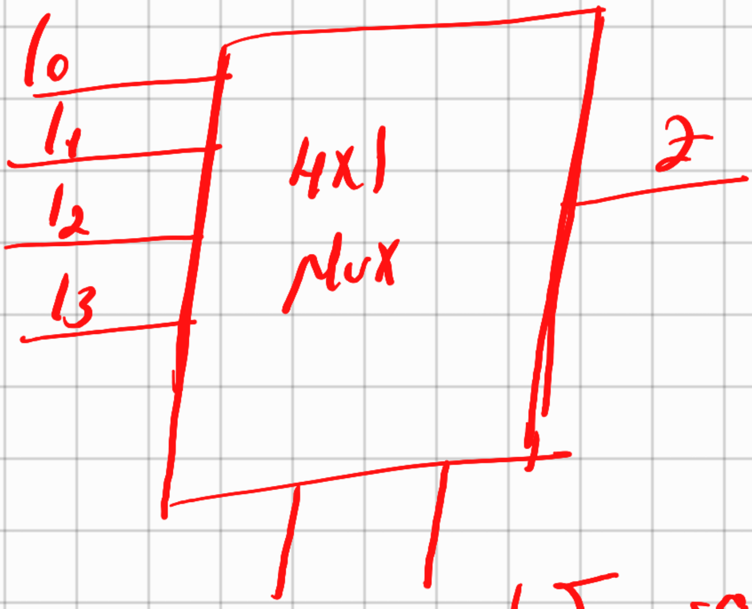
4- Seçiciler (Multiplexer-Mux)





S_1	S_0	Sayısal deger
0	0	0

0	0	0
0	1	1
1	0	2
1	1	3

Örnek → Bu devrede 4x1 Mux kullanarak f fonksiyonunu gerçekleyiniz

x_3, x_4 → Seçim uçları

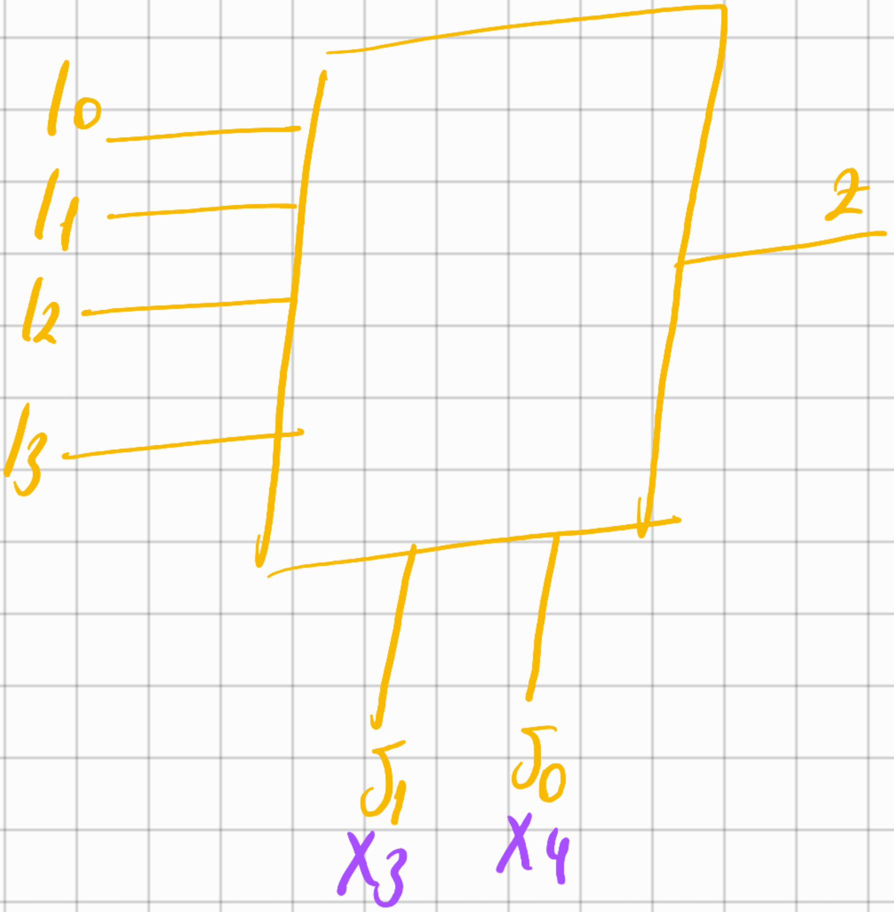
Örnek

x_1	x_2	x_3	x_4	$f(x_1, x_2, x_3, x_4)$
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	[illegible]	0	1	1

01		
01	10	0
01	11	1
10	00	1
10	01	1
10	10	1
10	11	1
11	00	0
11	01	0
11	10	0
11	11	0

$$\hookrightarrow f \rightarrow \bar{x}_1\bar{x}_2x_3\bar{x}_4 + \bar{x}_1\bar{x}_2x_3x_4 + \bar{x}_1x_2\bar{x}_3x_4 + \bar{x}_1x_2x_3x_4 + x_1\bar{x}_2\bar{x}_3\bar{x}_4 + x_1\bar{x}_2\bar{x}_3x_4 + x_1\bar{x}_2x_3\bar{x}_4 + x_1\bar{x}_2x_3x_4$$

↳ X_3 ve X_4 seçim ucları bunu kullanıca



$\bar{X}_3\bar{X}_4,\ X_3\bar{X}_4,\ \bar{X}_3X_4,\ X_3X_4$

Bunların parantezine alırız ifadeleri sonra kalan değerler artık input-ları bağlıyabilceğimiz duruma gelir

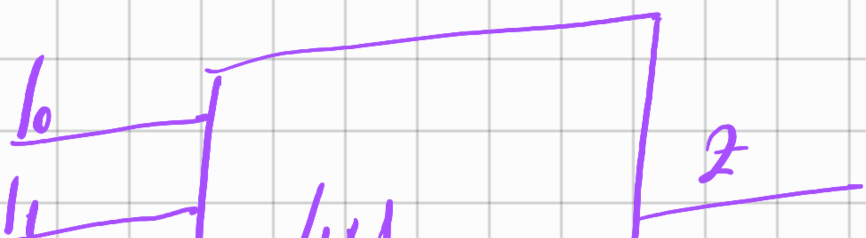
$$f \to \bar{X}_3\bar{X}_4(X_1\bar{X}_2) + \bar{X}_3X_4(\bar{X}_1X_2 + X_1\bar{X}_2)$$
$$+ X_3\bar{X}_4(\bar{X}_1\bar{X}_2 + X_1\bar{X}_2) + X_3X_4(\bar{X}_1\bar{X}_2 + \bar{X}_1X_2$$

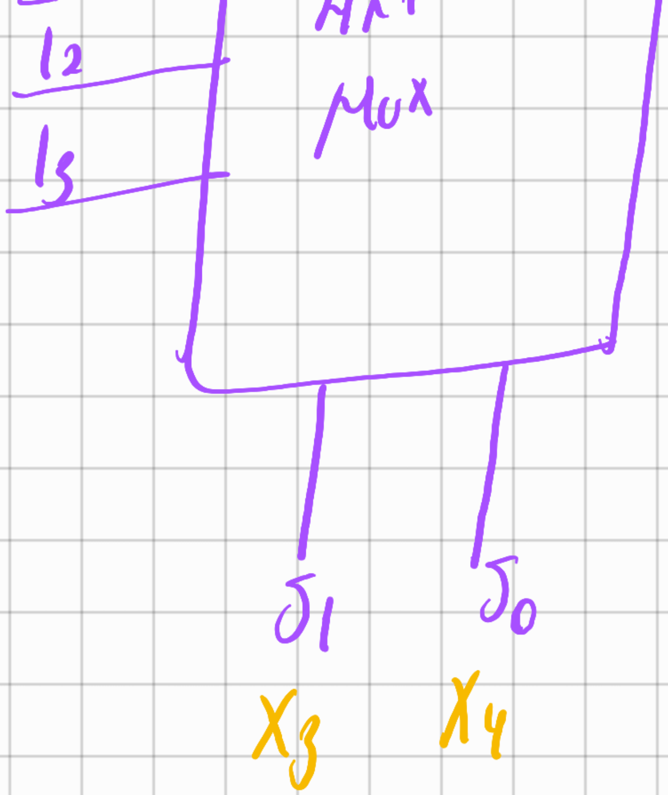
$+ x_3 x + (x_1 x_2$ $\quad + x_1 x_2)$

$$\hookrightarrow f \to \overset{0}{x_3'}\,\overset{0}{x_4'}\,(x_1\bar{x}_2) + \overset{0}{\bar{x}_3}\,\overset{1}{x_4}\,(x_1 \oplus x_2) + \overset{1}{x_3}\,\overset{0}{\bar{x}_4}\,(\bar{x}_2(\bar{x}_1 + x_1)) + \overset{1}{x_3}\,\overset{1}{x_4}\,\underbrace{(\bar{x}_1(\bar{x}_2 + x_2) + x_1\bar{x}_2)}_{\bar{x}_1 + (x_1\bar{x}_2)}$$

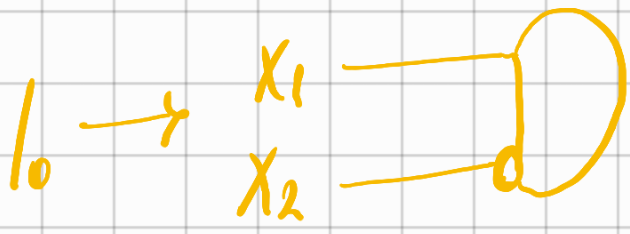
$$(\bar{x}_1 + x_1)(\bar{x}_1 + \bar{x}_2) \to (\bar{x}_1 + \bar{x}_2)$$

	S_1	S_0	
a	x_3	x_4	I_a
0	0	0	$x_1\bar{x}_2$
1	0	1	$x_1 \oplus x_2$
2	1	0	$\bar{x}_2$
3	1	1	$\bar{x}_1 + \bar{x}_2$

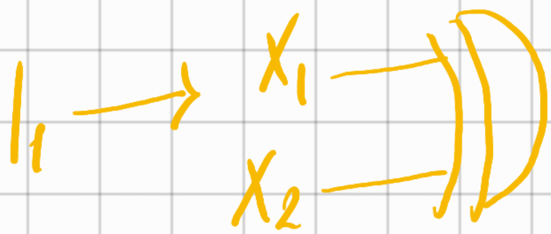




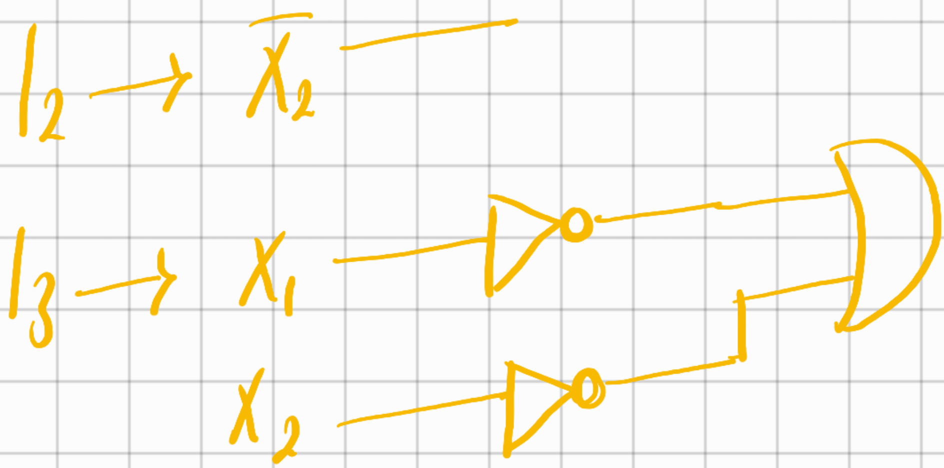
I2
I3
Mux
S1
S0
X3
X4



I0 →
X1
X2



I1 →
X1
X2



I2 →
$\bar{X}_2$
I3 →
X1
X2

Örnek 2 → Seçim yolarını X_3, X_4 yerine X_1 ve X_2 yaparsak ne olur

(Bu sefer X_1 ve X_2 üzerinden paranteze alır içerdeki ifadelere bakarız)

↳ $\bar{X}_1\bar{X}_2$, $\bar{X}_1X_2$, $X_1\bar{X}_2$, X_1X_2

↳ $f \rightarrow \bar{X}_1\bar{X}_2(X_3\bar{X}_4 + X_3 + X_4) + \bar{X}_1X_2(\bar{X}_3X_4 + X_3X_4) + X_1\bar{X}_2(\bar{X}_3\bar{X}_4 + \bar{X}_3X_4 + X_3\bar{X}_4 + X_3X_4) + X_1X_2(0)$

$\rightarrow \bar{X}_1\bar{X}_2(X_3(\cancel{\bar{X}_4 + X_4})_{1}) + \bar{X}_1X_2(X_4(\bar{X}_3 + X_3))$

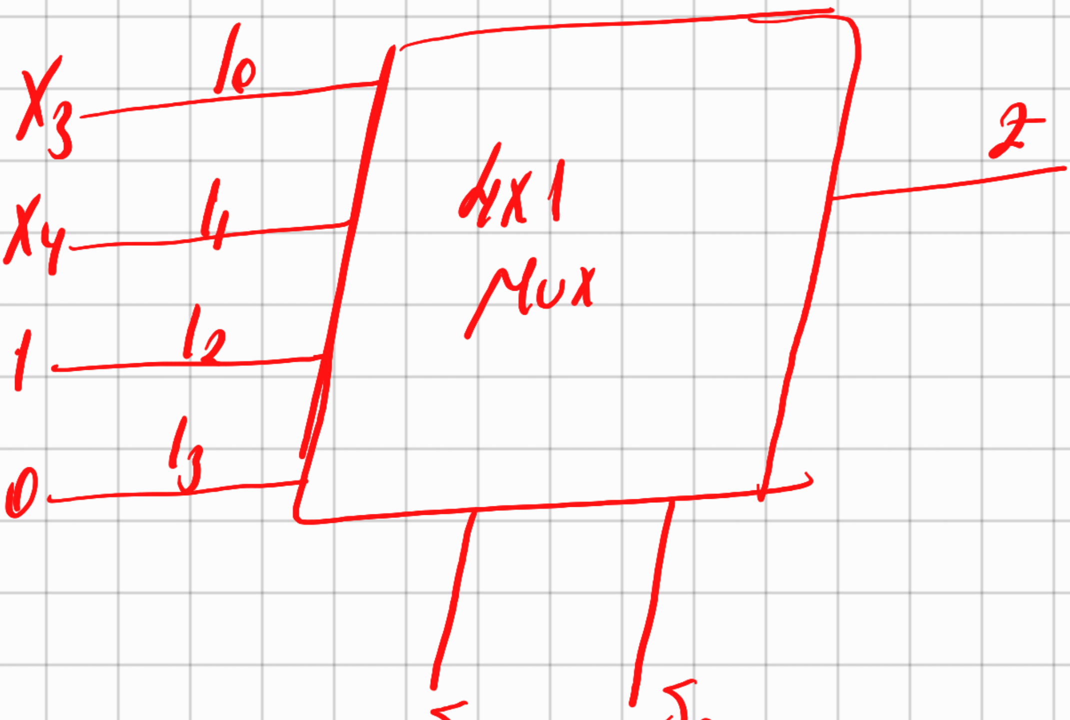
$X_1\bar{X}_2(\bar{X}_3(\bar{X}_4 + X_4) + X_3(\bar{X}_4 + X_4))$

$\cancel{\bar{X}_3 + X_3}$

1

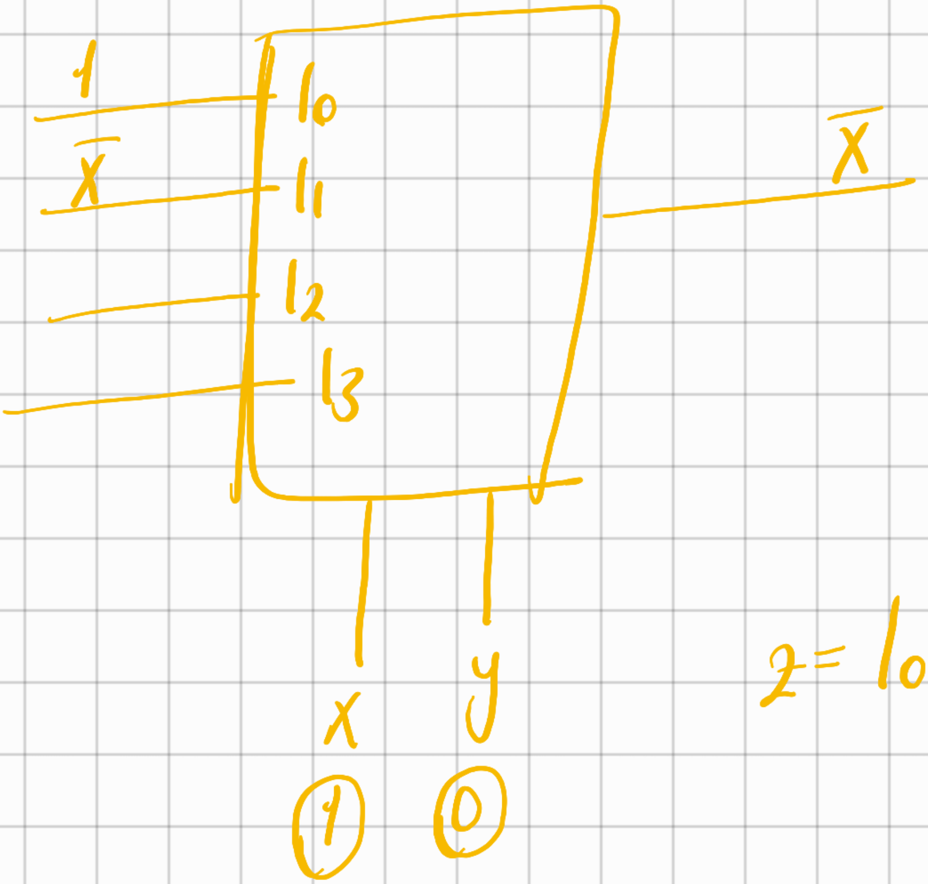
$\hookrightarrow \bar{X}_1\bar{X}_2X_3 + \bar{X}_1X_2X_4 + X\bar{X}_2 + X_1X_2(0)$

a	$X_1 X_2$	I_a
0	0 0	X_3
1	0 1	X_4
2	1 0	1
3	1 1	0



d_1 d_0

X_1 X_2



$(x.y).1$

$(x\bar{y})\bar{x}$

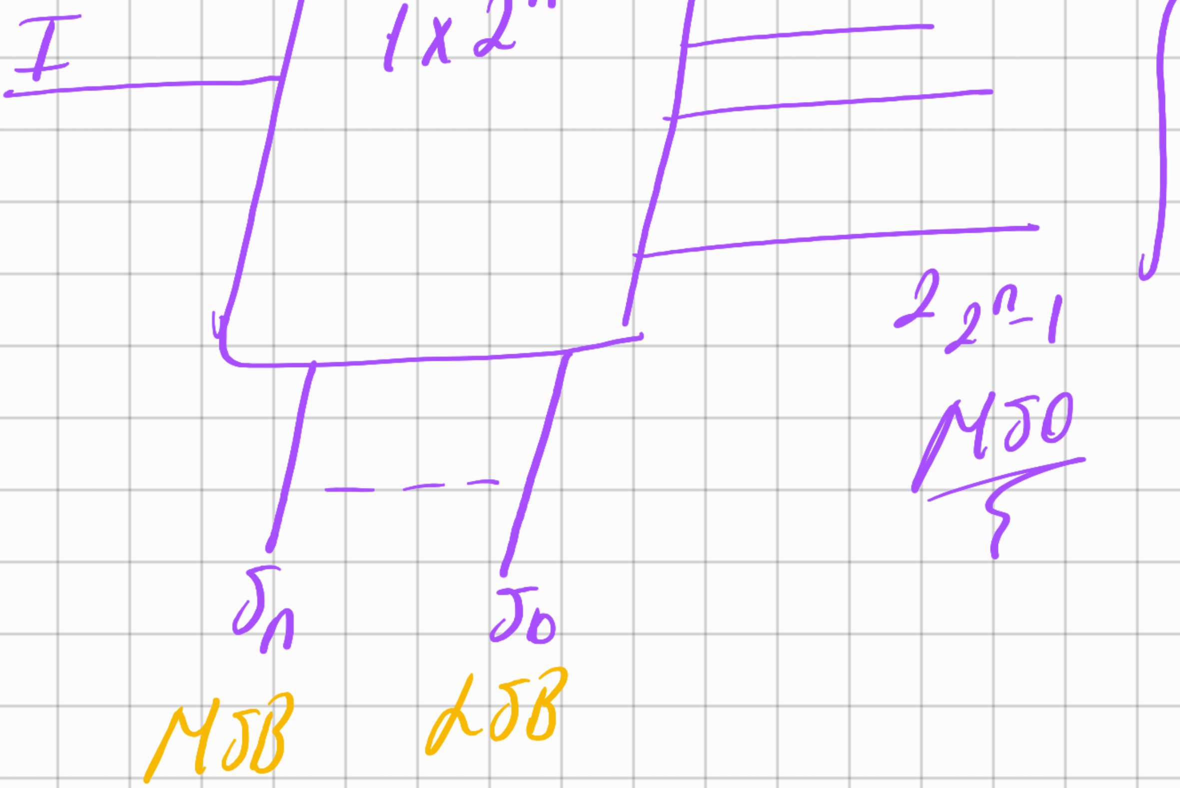
$$z = I_0(\bar{x}\bar{y}) + I_1(\bar{x}y) + I_2(x\bar{y}) + I_3(xy)$$

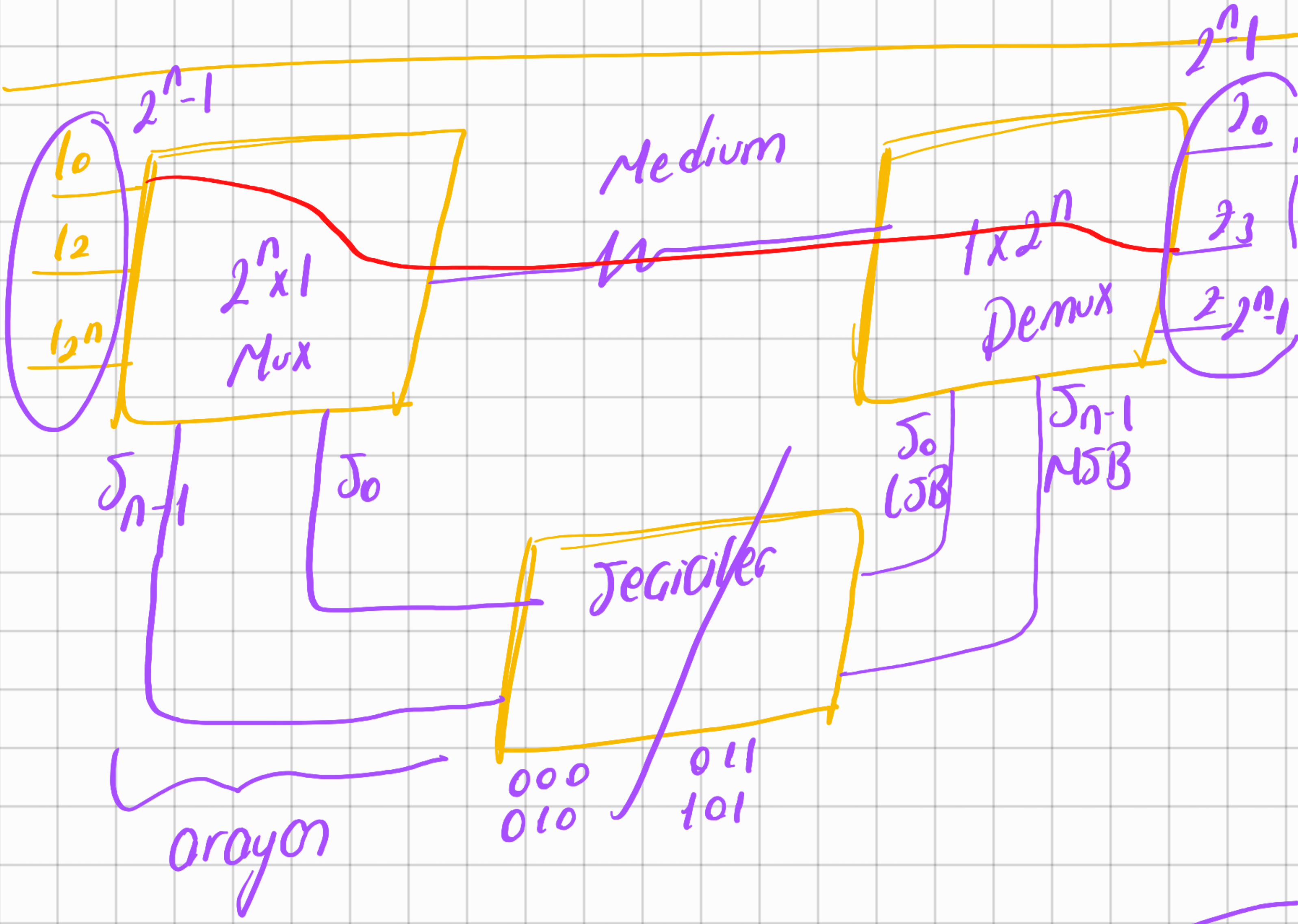
$x_1' + x_3$

↳ Sınavlarda böyle de çıkabilir

Dağıtıcılar (Demultiplexer - Demux)

2^n





Kod çözücü (Decoder) $n \times 2^n$

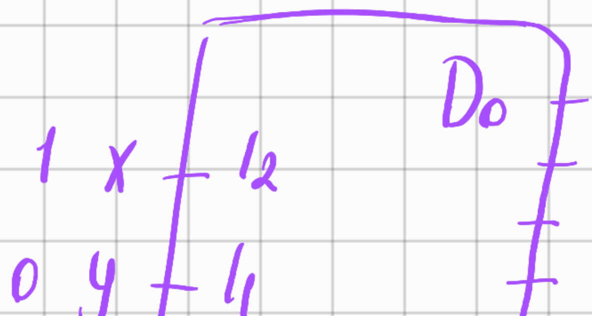
3×8

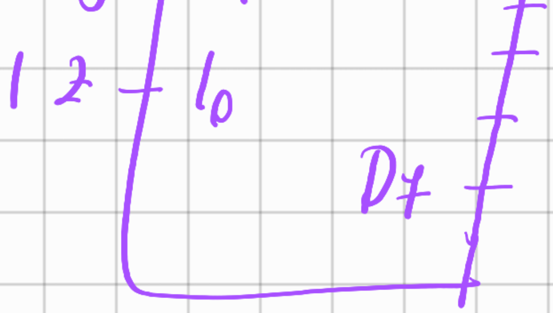
decoder
Mesela

↳ Sayısal değer

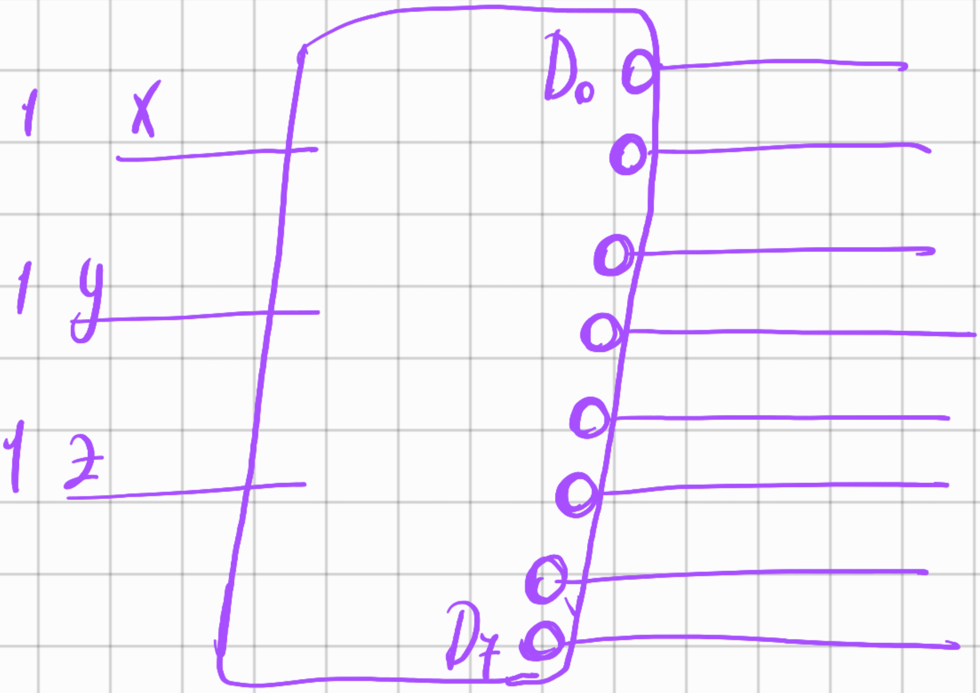
Sayısal değer	girişler $x\ y\ z$	çıkışlar $D_7\ D_6\ D_5\ D_4\ D_3\ D_2\ D_1\ D_0$
0	0 0 0	0 0 0 0 0 0 0 1
5	1 0 1	0 0 1 0 0 0 0 0
7	1 1 1	1 0 0 0 0 0 0 0
	1 1 1	0 1 1 1 1 1 1 1

tümleyen mantığında 7 böyle olur





Bu tümleyen mantığı ile çalışır



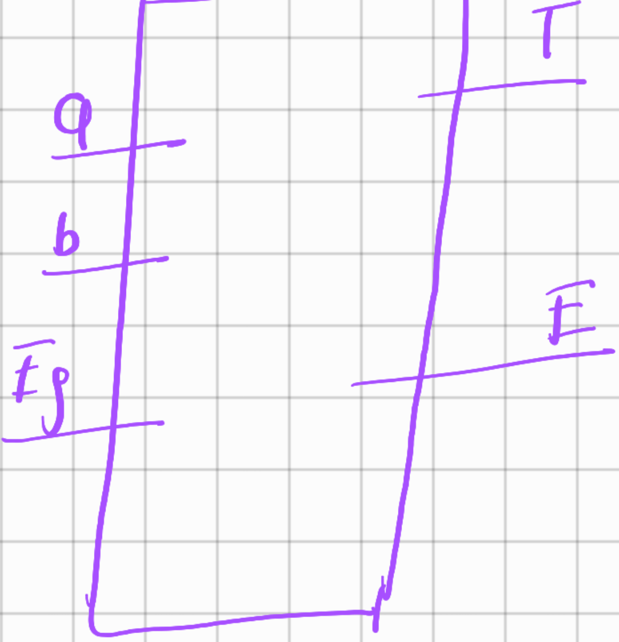
Örnek tam toplayıcı Devre

toplam çıkışı (a, b, E_g) =

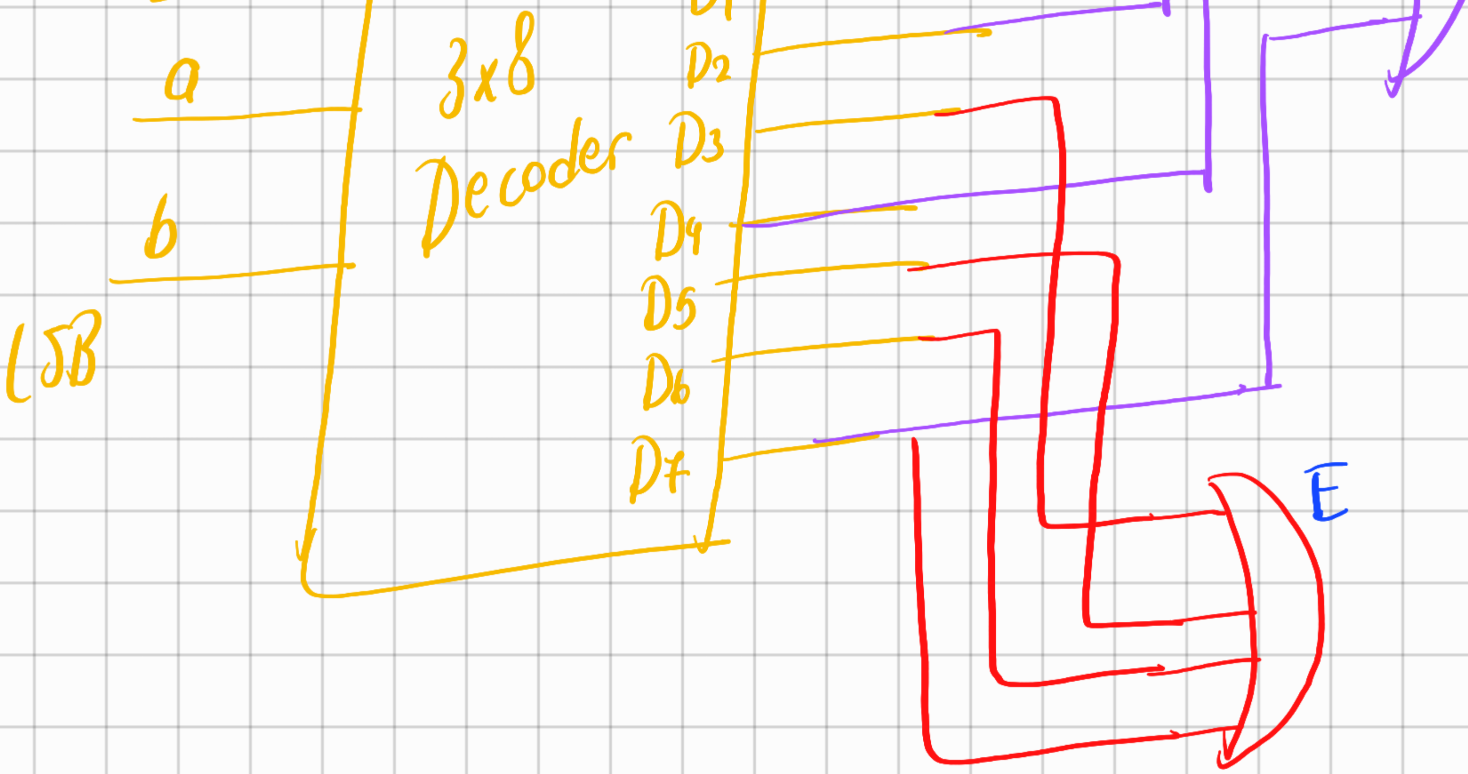
elde çıkışı (a, b, E_g)

$\sum(1,2,4,7)$'de toplam çıkışı 1 üretiyor

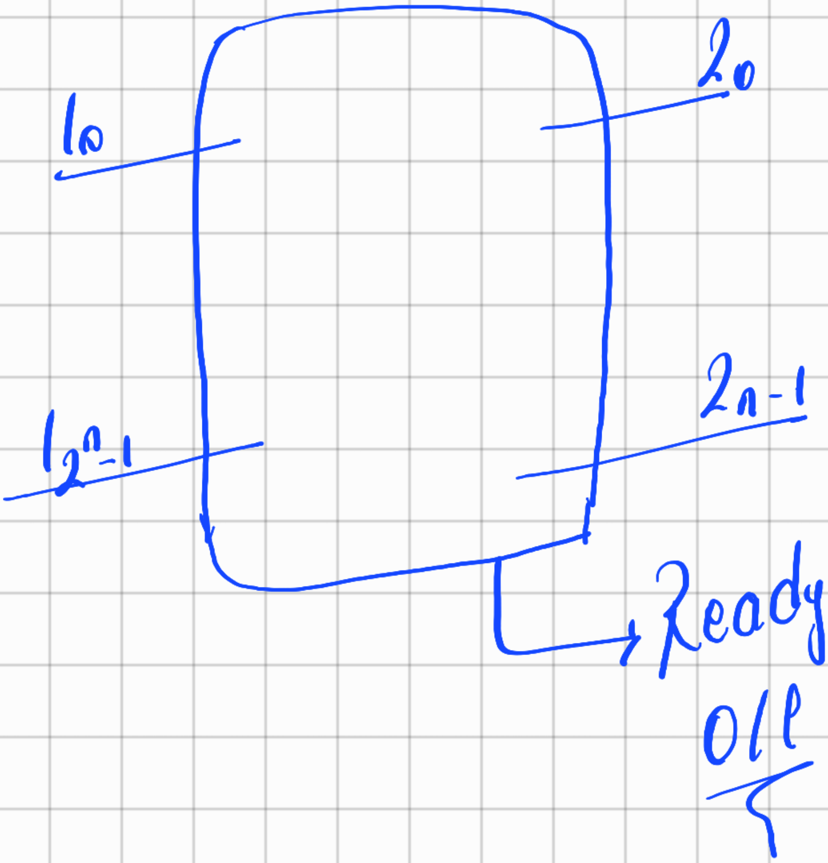
$\sum(3,5,6,7)$'de elde çıkışı 1 üretir



F_s	a	b	T	$\bar{E}$
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



Kodlayıcı (Encoder) $2^n \times n$

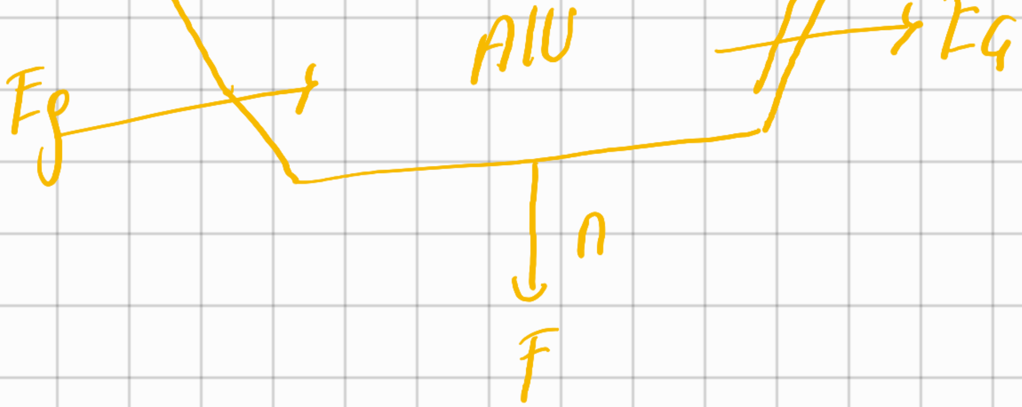


4x2 Encoder

Girişler (MSB I_3 … LSB I_0) $I_3\ I_2\ I_1\ I_0$	Çıkışlar $x\ y$	Hazır (ready)
0 0 0 0	∅ ∅	0
0 0 0 1	0 0	1
0 0 1 ∅	0 1	1
0 1 ∅ ∅	1 0	1
1 ∅ ∅ ∅	1 1	1

Bunlar ready'nin 1 olduğu yerler

Aritmetik logic Unit (ALU)



Örnek → Veri girişleri $\underline{A}$: $a_3a_2a_1a_0$ $\underline{B}$: $b_3b_2b_1b_0$

elde giriş çıkışları E_g / $E_ç$

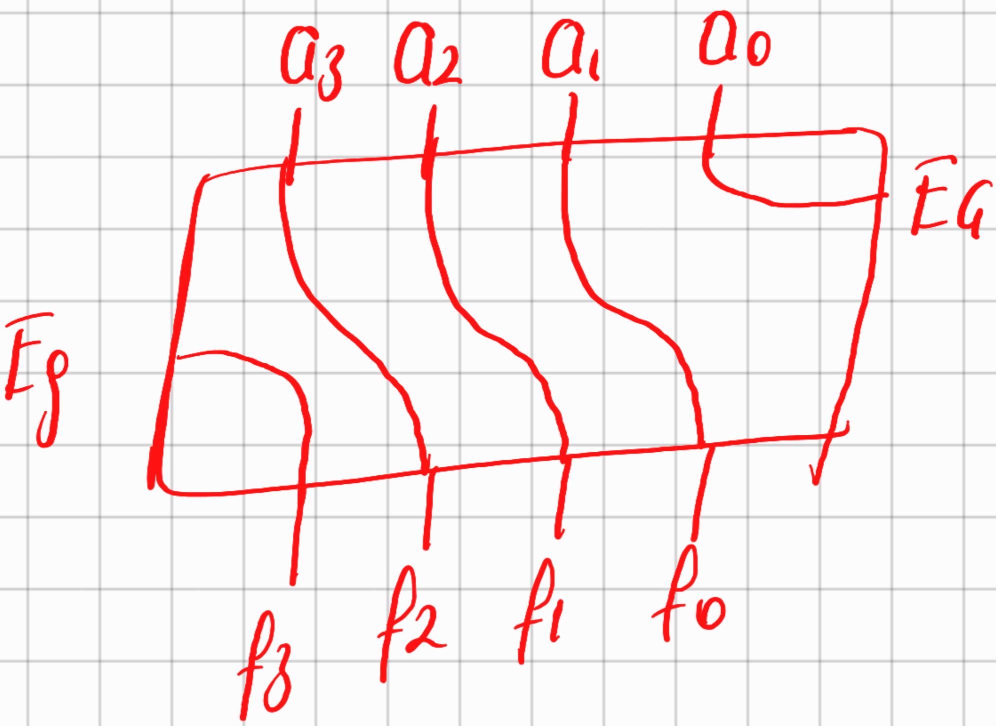
Sonuç $F(f_3f_2f_1f_0)$

işlem kodu $M(m_1m_0)$ → $2^M \to 2^2 = 4$ işlem tanımlanabilir

m_1m_0	işlem	
0 0	$F=A+B$	toplama TT (tam toplayıcı)
0 1	$F=A-B$	çıkarma TÇ (tam çıkarıcı)
		Complementi $(A)^c$

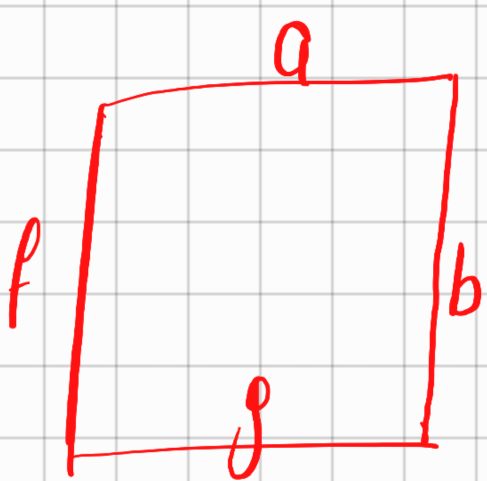
10 | $F=A$ | A'nın compl… (tümle

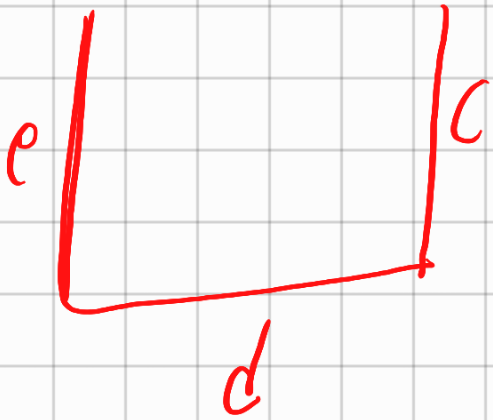
11 | $F=A_{1R}$ | A 1bit shif right (öteleyici) yici)

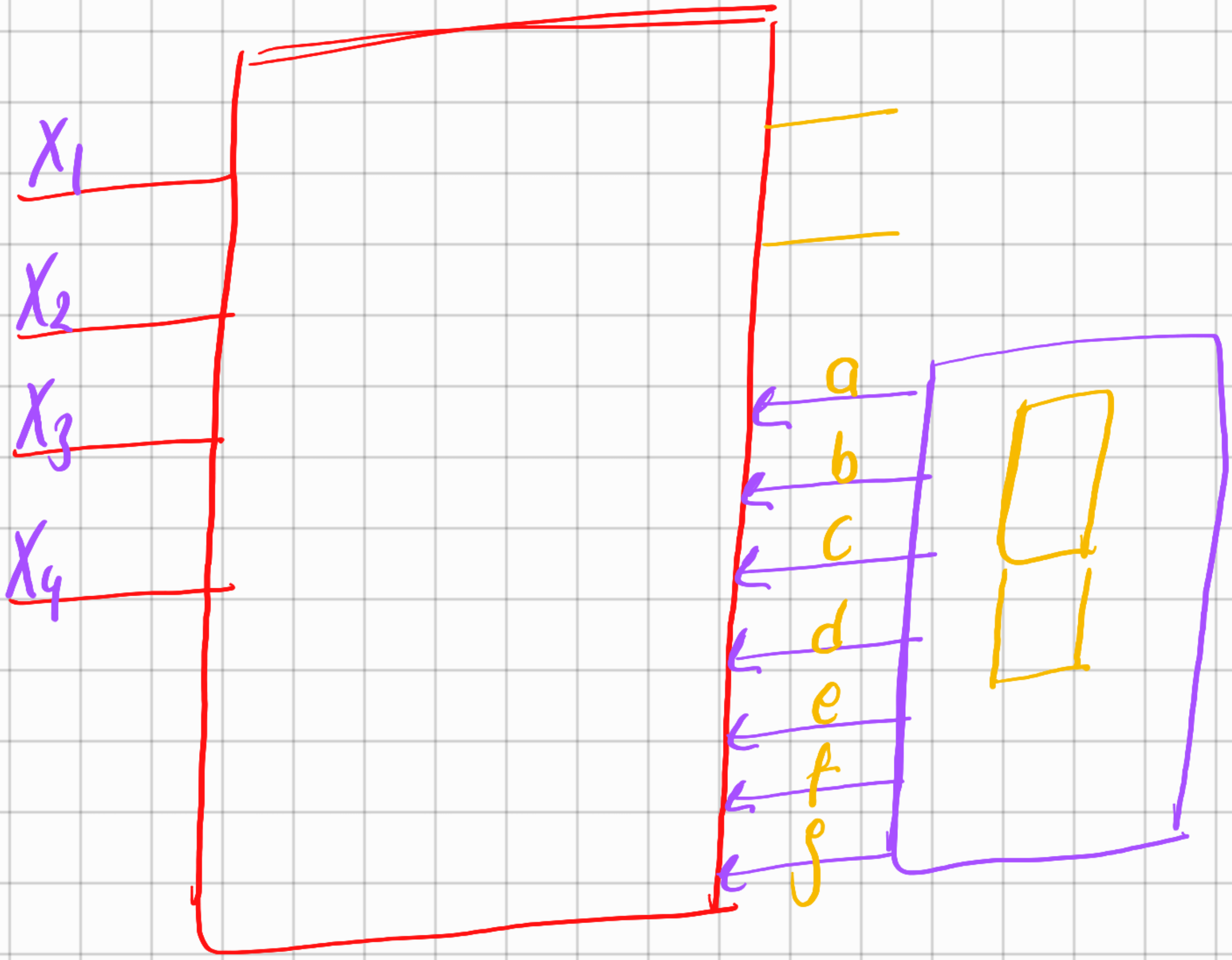


Burda bir şey var

7 segment display







10 sayı

$2^3 \rightarrow 8$

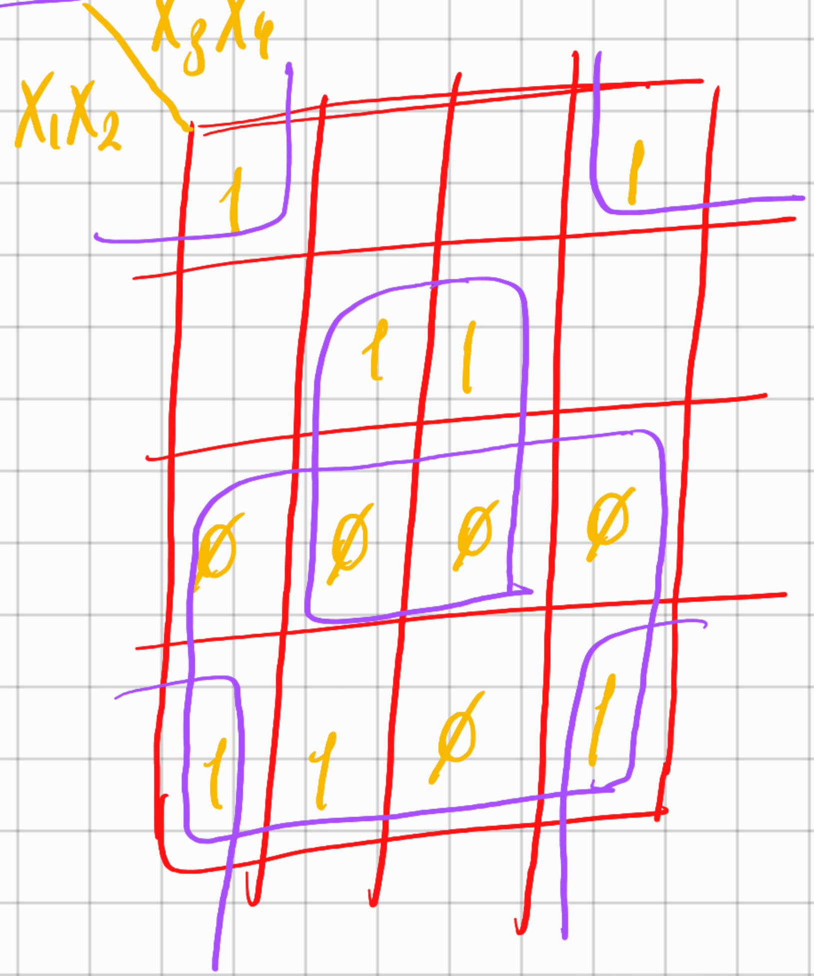
$2^4 \rightarrow 16$

↳ 6 boşa akar

$V_d > 0$
ise ışık verir

giriş					çıkışlar						
x_1	x_2	x_3	x_4	a	b	c	d	e	f	g
0	0	0	0	1	1	1	1	1	1	0
0	0	0	1	0	1	1	0	0	0	0
⋮										
1	0	0	1	1	1	1	1	0	1	1
1	0	1	0	∅	∅	∅	∅	∅	∅	∅
⋮										
1	1	1	1	∅	∅	∅	∅	∅	∅	∅

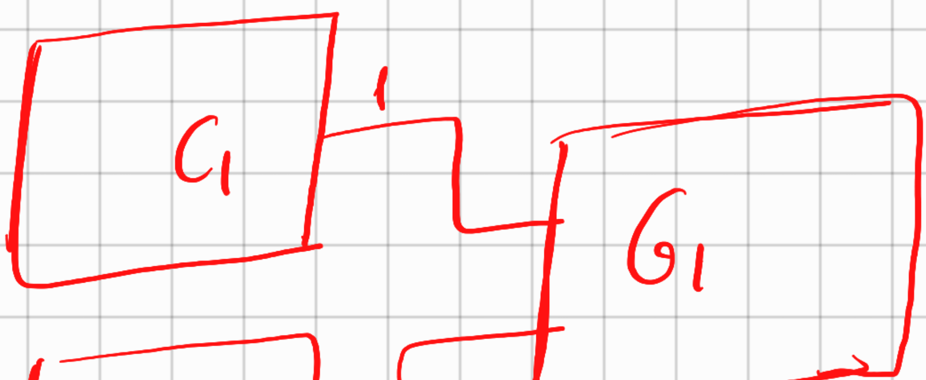
$\boxed{1001}$ → yazabileceğimiz en büyük sayı

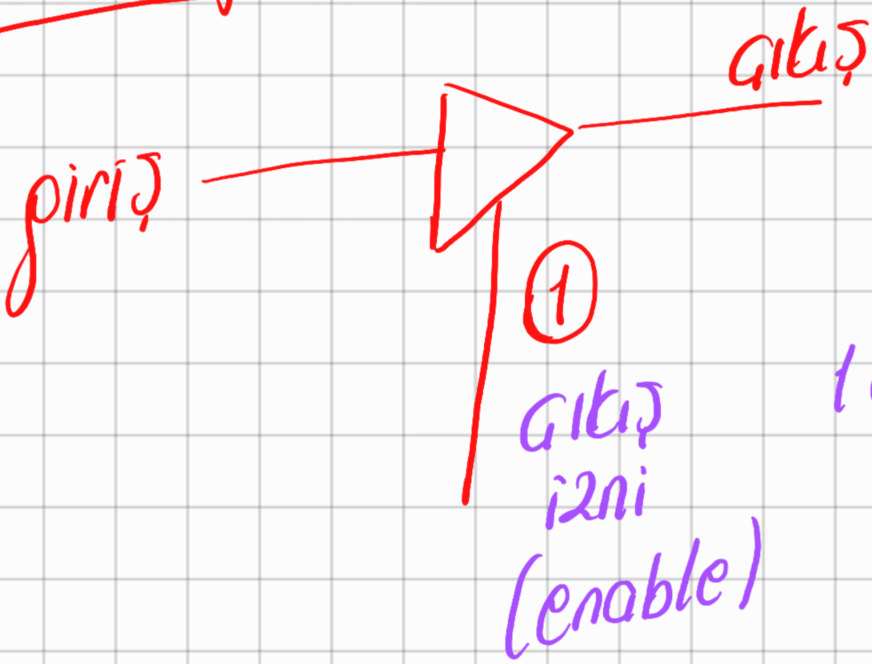


$\hookrightarrow a \rightarrow x_1 + \overline{x_2}\,\overline{x_4} + x_2 x_4$

$\hookrightarrow x_1 + x_2 \odot x_4$

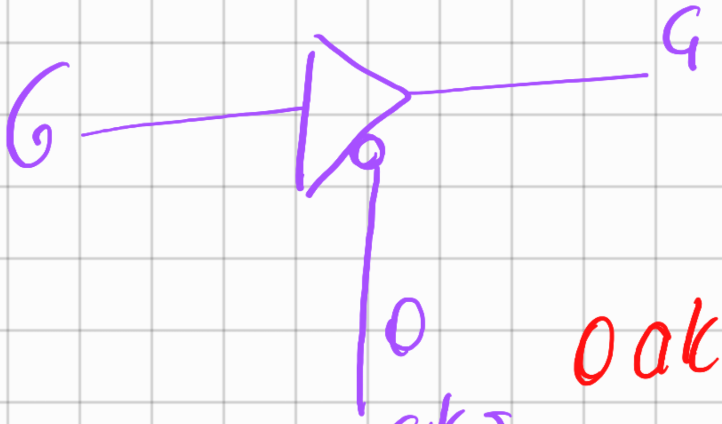
Beffer (tanpon)





1 aktif

çıkış izni	giriş	çıkış
0	0	YE
0	1	
1	0	0
1	1	1



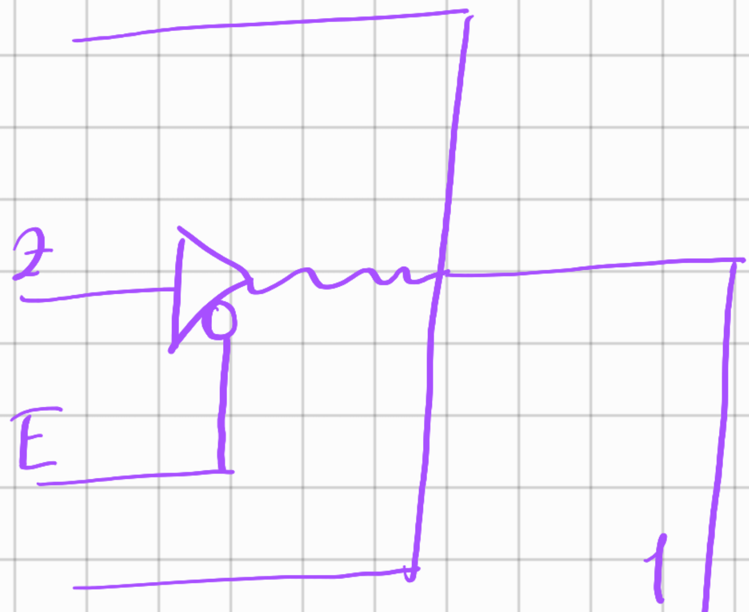
0 aktif

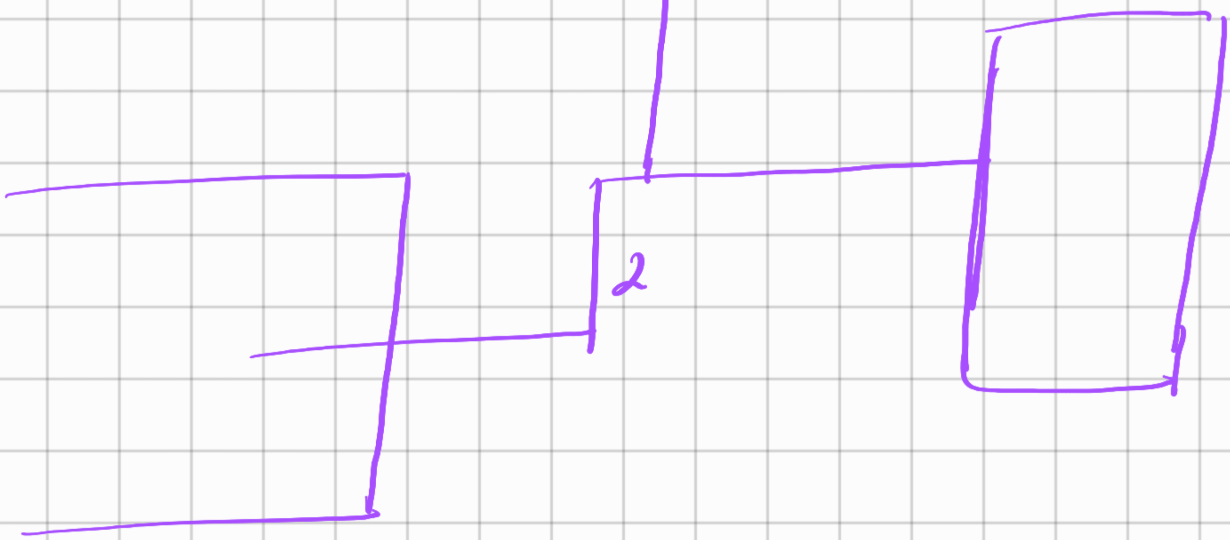
Çıkış
izni

Çıkış izni	giriş	çıkış
0	0	0
0	1	1
1	0	yüksek empedans
1	1	yüksek empedans

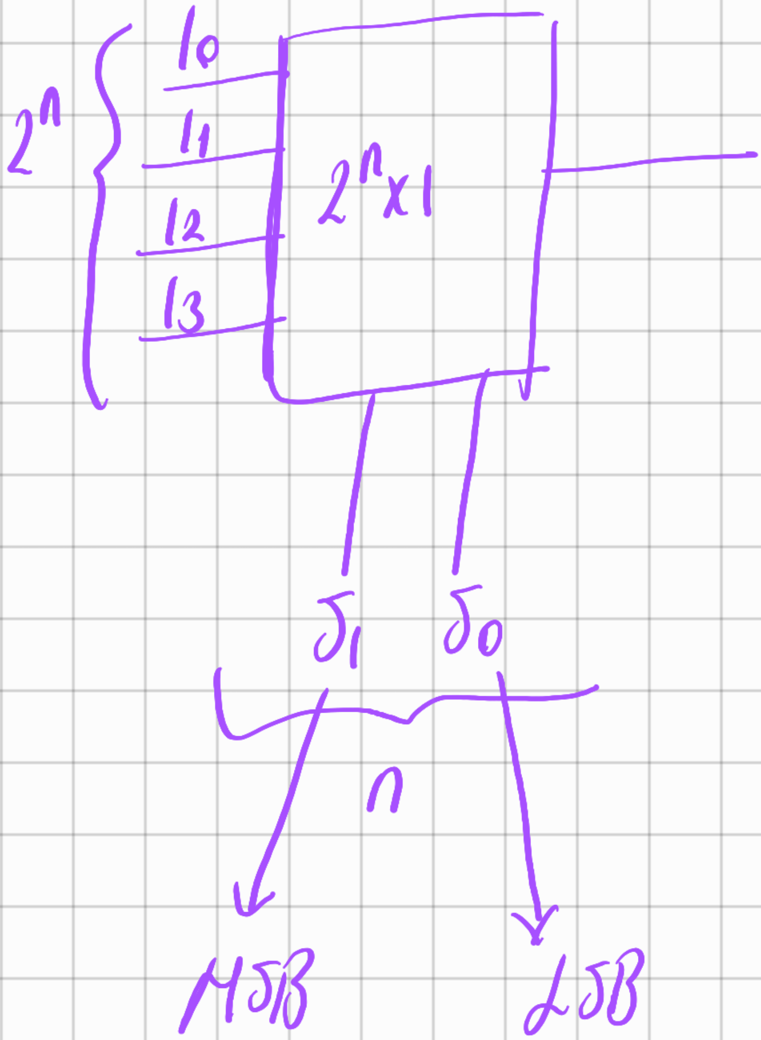
↳ yüksek empedans demek akım akmıycak demek

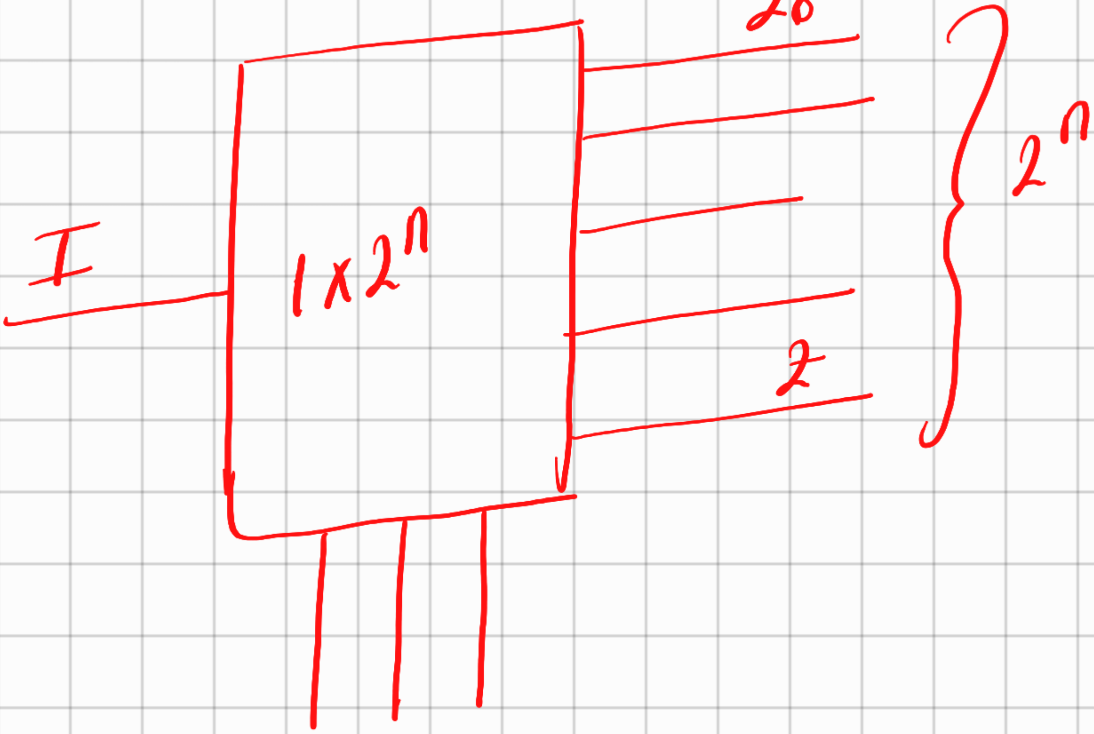
1- Bufferlar
2- Pull up direnci





Geçen Senenin Kayıtları





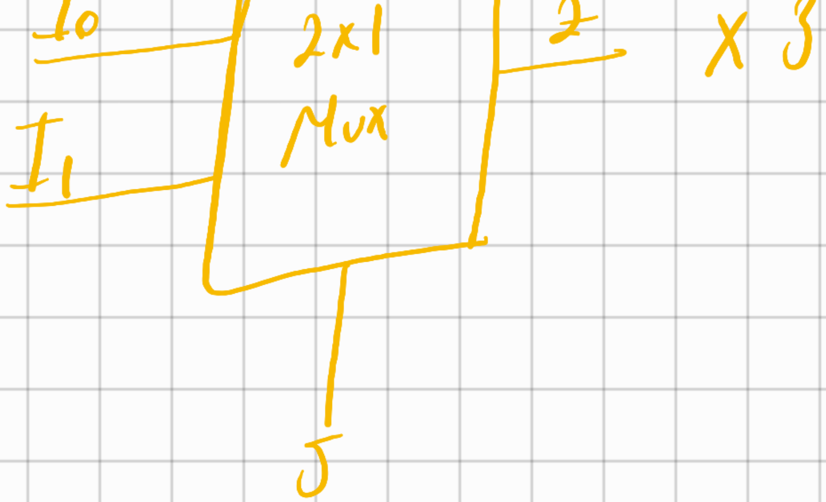
Örnek

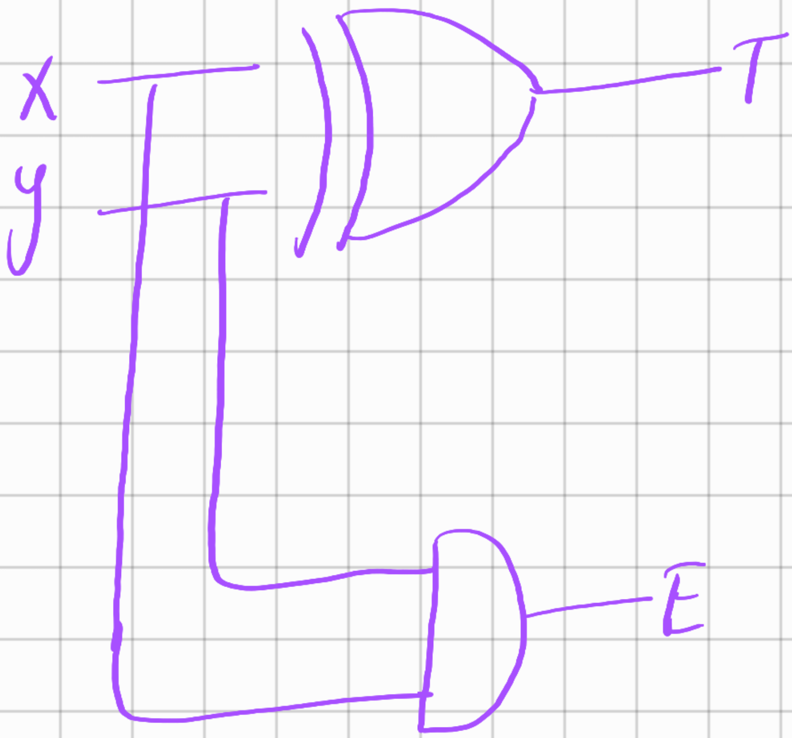
X	Y	T (Sum)	E (Carry)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

→ xy → yarı toplayıcı örneği

→ XOR

Örnek 3 tane 2x1 Mux kullanarak yarı toplayıcı gerçekleyiniz

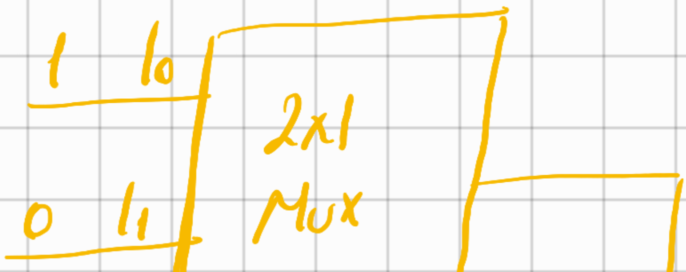


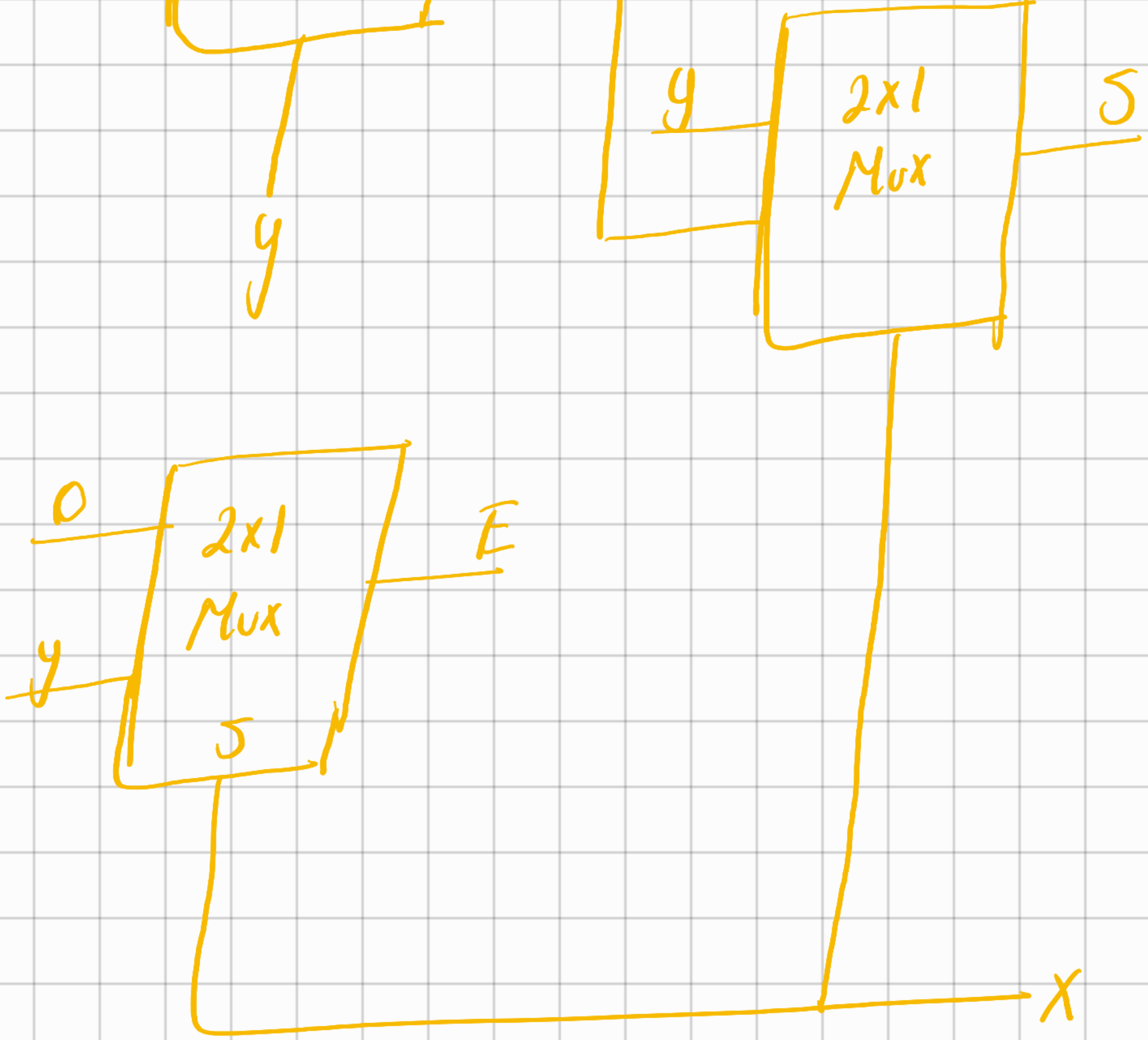


X
y
0
1

Çözüm

5



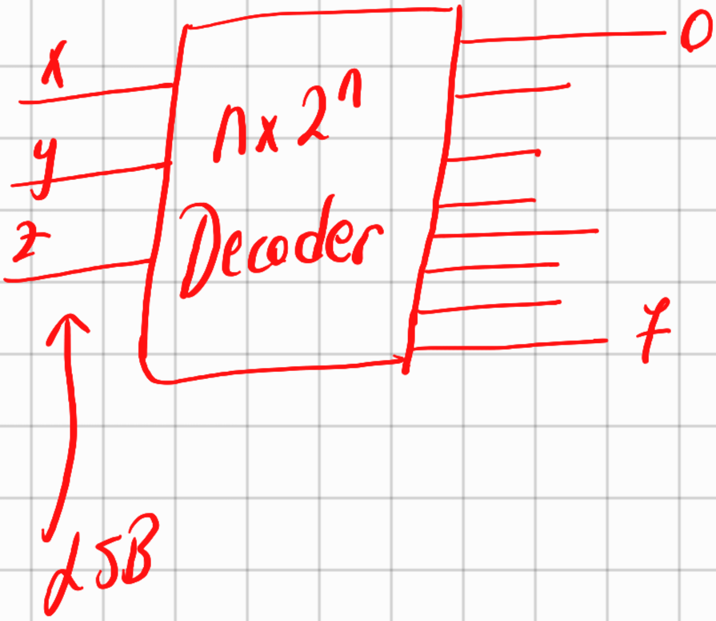


(4) Kod çözücü (Decoder)

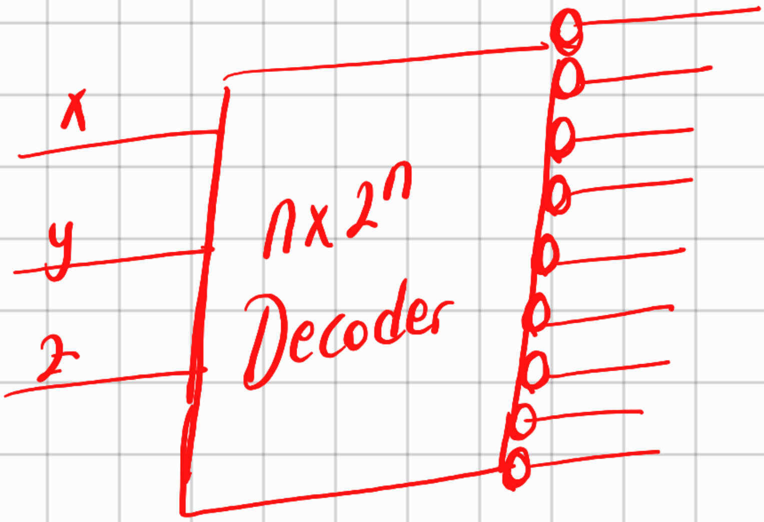
$n=3$

x	y	z	D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0

110 | $\underbrace{1\,0\,1\,1\,1\,1\,1\,1}_{\text{tümleyen çıkışlı}}$



1 aktif
0 pasif



1 pasif
0 aktif

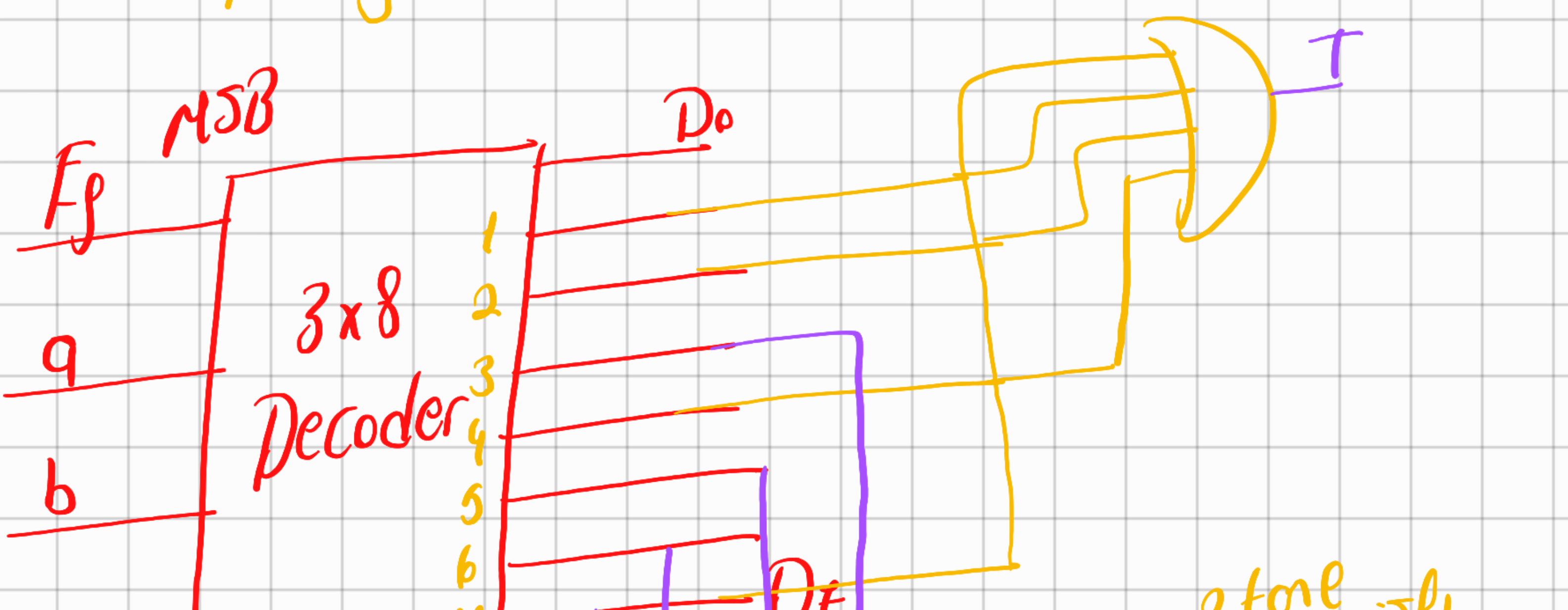
tümleyen çıkışlı

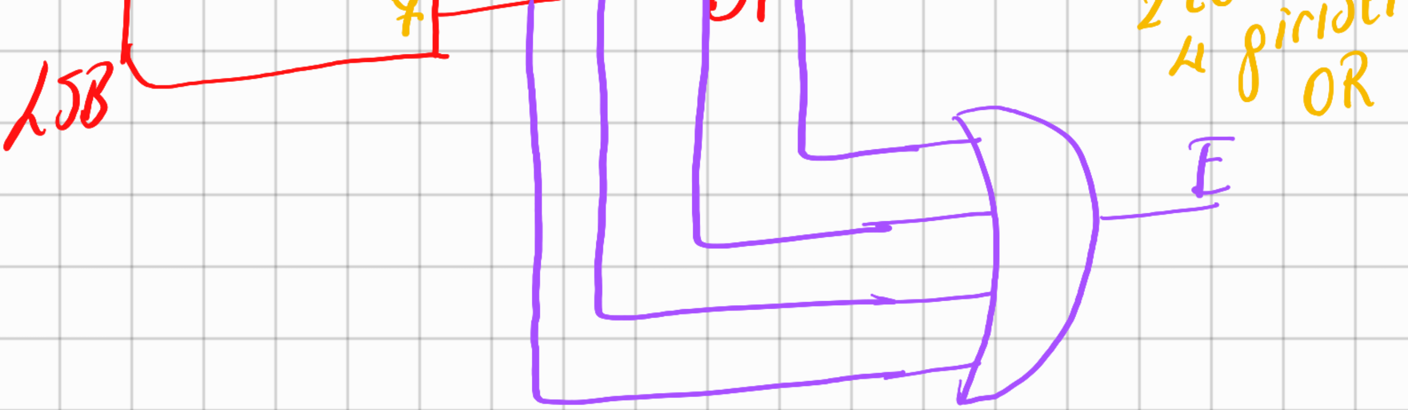
Ör Tam toplayıcıyı kod çözücü kullanarak tasarlayınız
S

E_p	a	b	T	E
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$$T(E_p, a, b) \rightarrow \sum(1,2,4,7)$$

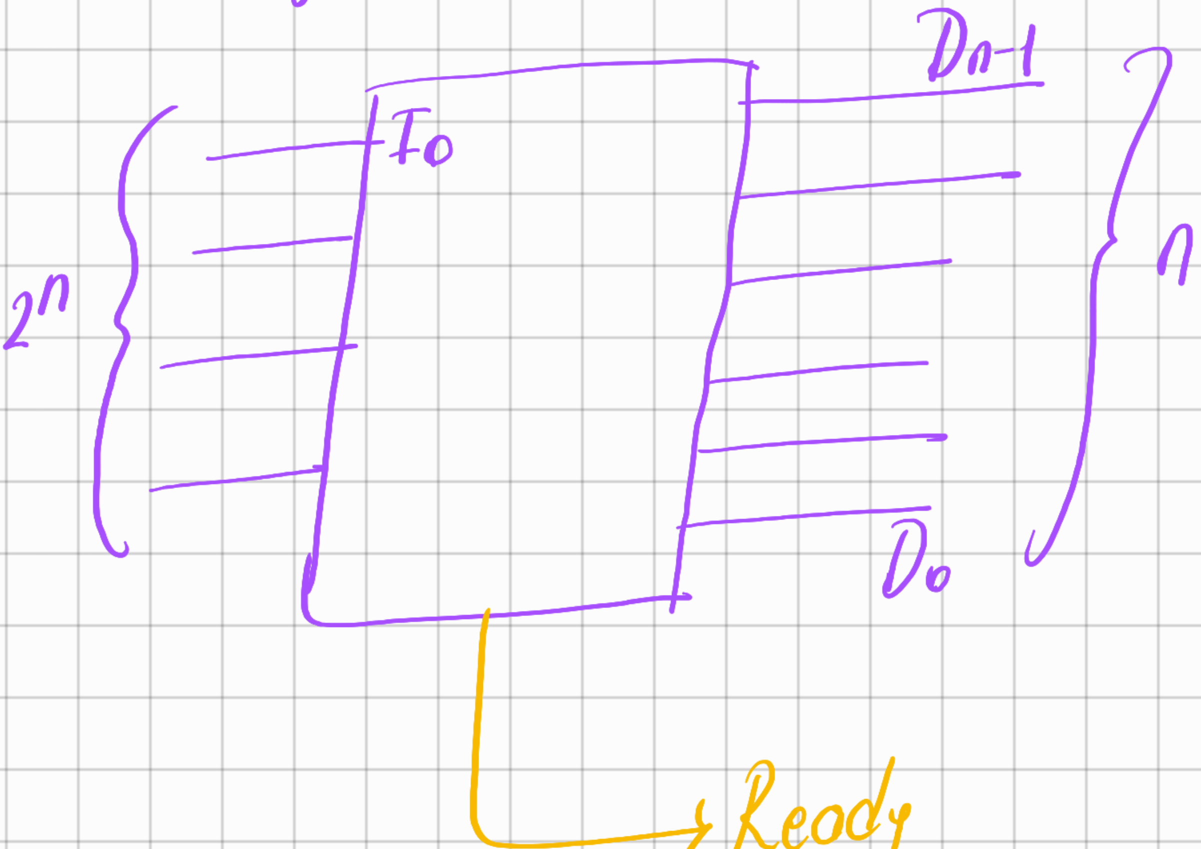
$$E(E_p, a, b) \rightarrow \sum(3,5,6,7)$$





⑤ Kodlayıcı (Encoder)

2^n giriş n çıkış



MSB girişler LSB | çıkışlar

I_3	I_2	I_1	I_0	D_1	D_2	Ready
0	0	0	0	Ø	Ø	0
0	0	0	1	0	0	1
0	0	1	Ø	0	1	1
0	1	Ø	Ø	1	0	1
1	Ø	Ø	Ø	1	1	1

6 ALU (Arithmetic logic UNIT)